COMP 018: Database Administration - Activity 6 Progress Report
Database Backup Planning, Verification, Restoration, and Disaster Recovery Simulation

Project Identification & Metadata

Course	: COMP 018 - Database Administration
Evaluating Group	: Group 4: Olivarez, Charles Kean E., Infantado, Gerwin G., Pocsidio, Jhon Louie F., Abangan, SJ Kaizen M., Bartolata, Redge Raindel N.
Evaluating Group Repository	: https://github.com/Nycri/brewpos
Target Project Monitored	: PUP-SPOT: Campus Lost and Found System
Target Project Group	: Group 5 (Imperial, Manansala, Napeñas, Lctz, Quody)
Target Project Repository	: https://github.com/grp05-dba/grp05_lostandfound
Primary Database	: grp05_lostandfounddb (PostgreSQL 18.6 on x86_64 Windows, compiled by MSVC)
Restored Test Database	: grp05_lostandfounddb_restore_test

ACTIVITY 6: Database Backup, Restoration, and Recovery

3.1 Backup Planning & Architecture
The disaster recovery plan for the PUP-SPOT database encompasses eight relational tables within the public schema, capturing category definitions, cataloged found items, student lost reports, filed claim requests, match scores, and image metadata. The plan covers sequence counters, primary keys, foreign keys, constraints, and privilege grants to ensure operational continuity.

The backup strategy utilizes PostgreSQL custom compressed archive format via a dedicated backup operator role. Archive dumps are stored in a protected directory outside web roots and repository trees. Recovery objectives define a Recovery Point Objective of 1 hour and a Recovery Time Objective of 30 minutes. Retention guidelines specify maintaining seven daily backups alongside four weekly snapshots.

Because database login accounts exist at the PostgreSQL cluster level and are not stored within single-database dumps, disaster recovery plans require executing access-control role scripts on new server instances prior to restoring database contents.

3.2 Backup Creation, Checksum & Archive Inspection
Logical backup creation was executed using the dedicated backup account, producing a compressed custom format archive sized at 35,083 bytes. Cryptographic verification computed a SHA-256 hash value of FA6808CEA3E2ACAF86A8887B5BAF86E0444A6C5A8C899D9B3D75F9EBB02D352C, establishing a baseline for integrity verification.

Inspection of the archive table of contents verified 91 catalog entries, confirming the inclusion of

all eight database tables, sequence objects, primary keys, check constraints, foreign keys, and default privilege grants. To ensure recoverability beyond file inspection checks, a complete restoration drill was conducted in an isolated test database.

3.3 Restoration into Separate Test Database
An isolated test database named grp05_lostandfounddb_restore_test was created on the local server instance. The custom archive dump file was restored into this database using administrative utility tools. The restoration process completed cleanly with zero error codes.

3.4 Post-Restore Verification Suite
Post-restoration verification confirmed that all eight relational tables were recreated with proper schema structure under administrative ownership. Record count validation across tables demonstrated complete parity with the source database, matching eleven categories, seven items, eight lost reports, three claims, and six potential matches. Database checks confirmed that all 53 constraints were fully restored.

Access control verification on the restored test database confirmed that privilege definitions transferred accurately. Queries executed under the public user account successfully returned item records, while update attempts on catalog items were blocked by the database security layer.

3.5 Controlled Recovery Simulation
A controlled data recovery simulation was performed to test incident response procedures. A disposable item record was inserted into the restored test database to establish a baseline state. A data loss event was then simulated by removing the record, and queries confirmed its deletion.

Targeted record restoration was executed to reinsert the missing data row without affecting existing database entries. Post-recovery verification confirmed that the record was fully restored with its original attributes intact. Following verification, the test record was removed to clean up the test environment.

3.6 Limitations, Evaluation & Production Recommendations
Restoration procedures on fresh server installations require pre-executing role definition scripts to establish user accounts before restoring database ACLs. Because image tables reference file storage paths, database backups must be paired with filesystem backups of storage directories. For production operations, daily automated backup generation should be scheduled using system task schedulers to archive database dumps with date-stamped filenames and retain snapshots according to policy. Enabling continuous write-ahead log archiving will support point-in-time recovery capabilities.

CLASS GOOGLE SHEETS TRACKER ENTRY (Activity 6)

Column Header	Value
Project	Group 5 - PUP-SPOT: Campus Lost and Found System
Database and backup fil	grp05_lostandfounddb, grp05_lostandfounddb_activity6.dump, Size: 35,083 bytes, SHA256: FA6808CEA3E2ACAF86A8887B5BAF86E0444A6C5A8C899D9B3D75F9EBB02D352C
Backup method	pg_dump custom format (-Fc), compressed binary dump (2026-11-05 04:46:41)
Restore test result	Success (Restored into grp05_lostandfounddb_restore_test; all 8 tables, constraints, and data verified)
Access control after restore	Yes (Activity 4 roles and permissions validated; lostandfound_public denied UPDATE on items)
Recovery simulation result	Recovered (Simulated deleted test item successfully restored from verified baseline state)
Proposed RPO and RTO	RPO: 1 hour, RTO: 30 minutes
Evidence folder link	https://github.com/Nycri/brewpos/tree/main/activity-6/Evidence-act6
Feedback sent to owners	Yes (05/11/2026)
PDF progress document	Activity6_Progress_Report_Group5.pdf
Last updated	2026-11-05

AI USAGE DECLARATION

Google Gemini 3.8 Flash was utilized during project evaluation to inspect database schema definitions, evaluate relational hierarchy structures, formulate role verification test scenarios, analyze database maintenance operations and query performance execution plans, verify backup creation and restoration procedures, and structure technical report documentation. All

database statements, configuration checks, and verification procedures were directly executed and validated against PostgreSQL 18.6 within the local database environment.